

Problem Sheet: Chain Rule and Substitutions 1

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Exercise 1. Differentiate the following functions.

1. $\ln(5x)$
2. e^{-20x^2}
3. $\sin(16x^3)$
4. $\cos\left(\frac{2}{x}\right)$

Exercise 2. Evaluate the following indefinite integrals.

1. $\int \sin^2 x \, dx$
2. $\int \cos^2 x \, dx$

Exercise 3. For each integral below, state a suitable substitution before evaluating the indefinite integral.

1. $\int \frac{5}{5x+2} \, dx$
2. $\int e^{-20x} \, dx$
3. $\int \sin(16x) \, dx$
4. $\int \cos(2x) \, dx$